

In [1]:

```
pip install pandas
```

Requirement already satisfied: pandas in c:\users\prasa\anaconda3\lib\site-packages (2.3.3)
Requirement already satisfied: numpy>=1.26.0 in c:\users\prasa\anaconda3\lib\site-packages (from pandas) (2.3.5)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\prasa\anaconda3\lib\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\prasa\anaconda3\lib\site-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in c:\users\prasa\anaconda3\lib\site-packages (from pandas) (2025.2)
Requirement already satisfied: six>=1.5 in c:\users\prasa\anaconda3\lib\site-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
Note: you may need to restart the kernel to use updated packages.

In [2]:

```
import pandas as pd  
data=pd.read_csv(r"C:\Users\prasa\Downloads\trends.csv")
```

In [15]:

```
data.head()
```

	location	year	category	rank	query
0	Global	2001	Consumer Brands	1	Nokia
1	Global	2001	Consumer Brands	2	Sony
2	Global	2001	Consumer Brands	3	BMW
3	Global	2001	Consumer Brands	4	Palm
4	Global	2001	Consumer Brands	5	Adobe

In [4]:

```
data.head(50)
```

	location	year	category	rank	query
0	Global	2001	Consumer Brands	1	Nokia
1	Global	2001	Consumer Brands	2	Sony
2	Global	2001	Consumer Brands	3	BMW
3	Global	2001	Consumer Brands	4	Palm
4	Global	2001	Consumer Brands	5	Adobe
5	Global	2001	Men	1	Nostradamus
6	Global	2001	Men	2	Osama bin Laden
7	Global	2001	Men	3	Eminem
8	Global	2001	Men	4	Michael Jackson
9	Global	2001	Men	5	Howard Stern
10	Global	2001	Movies	1	Harry Potter
11	Global	2001	Movies	2	Lord of the Rings
12	Global	2001	Movies	3	Final Fantasy
13	Global	2001	Movies	4	Tomb Raider
14	Global	2001	Movies	5	Shrek
15	Global	2001	MP3 Music Services	1	Morpheus
16	Global	2001	MP3 Music Services	2	Napster
17	Global	2001	MP3 Music Services	3	Gnutella
18	Global	2001	MP3 Music Services	4	Kazaa
19	Global	2001	MP3 Music Services	5	Audiogalaxy
20	Global	2001	Music Groups	1	Beatles
21	Global	2001	Music Groups	2	U2
22	Global	2001	Music Groups	3	N Sync
23	Global	2001	Music Groups	4	Backstreet Boys
24	Global	2001	Music Groups	5	Linkin Park
25	Global	2001	New Products	1	Windows XP
26	Global	2001	New Products	2	Xbox
27	Global	2001	New Products	3	PlayStation 2
28	Global	2001	New Products	4	GameCube
29	Global	2001	New Products	5	Nokia 5510
30	Global	2001	News Resources	1	CNN

	location	year	category	rank	query
31	Global	2001	News Resources	2	BBC
32	Global	2001	News Resources	3	New York Times
33	Global	2001	News Resources	4	MSNBC
34	Global	2001	News Resources	5	Washington Post
35	Global	2001	Retailers	1	Amazon
36	Global	2001	Retailers	2	IKEA
37	Global	2001	Retailers	3	Toys "R" Us
38	Global	2001	Retailers	4	Walmart
39	Global	2001	Retailers	5	Circuit City
40	Global	2001	Sports	1	Anna Kournikova
41	Global	2001	Sports	2	Tour de France
42	Global	2001	Sports	3	Dale Earnhardt
43	Global	2001	Sports	4	Wimbledon
44	Global	2001	Sports	5	US Open
45	Global	2001	TV Shows	1	Loft Story
46	Global	2001	TV Shows	2	Big Brother
47	Global	2001	TV Shows	3	The Simpsons
48	Global	2001	TV Shows	4	Survivor
49	Global	2001	TV Shows	5	Gran Hermano

In [5]:

```
data
```

	location	year	category	rank	query
0	Global	2001	Consumer Brands	1	Nokia
1	Global	2001	Consumer Brands	2	Sony
2	Global	2001	Consumer Brands	3	BMW
3	Global	2001	Consumer Brands	4	Palm
4	Global	2001	Consumer Brands	5	Adobe
...	...	...	...	...	...
26950	Vietnam	2020	Là Gì?	1	Virus Corona là gì
26951	Vietnam	2020	Là Gì?	2	Miễn thị thực là gì
26952	Vietnam	2020	Là Gì?	3	Đầu cắt moi là gì
26953	Vietnam	2020	Là Gì?	4	Bệnh bạch hầu là gì
26954	Vietnam	2020	Là Gì?	5	Đông Lào là gì

26955 rows × 5 columns

In [6]:

```
data.tail()
```

	location	year	category	rank	query
26950	Vietnam	2020	Là Gì?	1	Virus Corona là gì
26951	Vietnam	2020	Là Gì?	2	Miễn thị thực là gì
26952	Vietnam	2020	Là Gì?	3	Đầu cắt moi là gì
26953	Vietnam	2020	Là Gì?	4	Bệnh bạch hầu là gì
26954	Vietnam	2020	Là Gì?	5	Đông Lào là gì

In [7]:

```
data.shape
```

(26955, 5)

In [8]:

```
data.info
```

```
<bound method DataFrame.info of      location  year      category  rank      query
0      Global  2001  Consumer Brands    1      Nokia
1      Global  2001  Consumer Brands    2        Sony
2      Global  2001  Consumer Brands    3         BMW
3      Global  2001  Consumer Brands    4         Palm
```

```
4      Global  2001  Consumer Brands      5      Adobe
...      ...    ...      ...      ...
26950 Vietnam  2020      Là Gì?      1  Virus Corona là gì
26951 Vietnam  2020      Là Gì?      2  Miễn thị thực là gì
26952 Vietnam  2020      Là Gì?      3  Đầu cắt moi là gì
26953 Vietnam  2020      Là Gì?      4  Bệnh bạch hầu là gì
26954 Vietnam  2020      Là Gì?      5      Đông Lào là gì
```

[26955 rows x 5 columns]>

In [9]:

```
data.describe()
```

	year	rank
count	26955.000000	26955.00000
mean	2015.243369	3.00000
std	3.564683	1.41424
min	2001.000000	1.00000
25%	2013.000000	2.00000
50%	2016.000000	3.00000
75%	2018.000000	4.00000
max	2020.000000	5.00000

In [10]:

```
data.columns
```

Index(['location', 'year', 'category', 'rank', 'query'], dtype='object')

In [11]:

```
data.dtypes
```

```
location    object
year        int64
category    object
rank        int64
query       object
dtype: object
```

In [12]:

```
data.isnull().sum()
```

```
location    0
year        0
category    0
rank        0
query       0
dtype: int64
```

In [13]:

```
data.duplicated().sum()
```

np.int64(10)

In [14]:

```
data.nunique()
```

```
location      83
year          20
category     2450
rank          5
query       18431
dtype: int64
```